University of Illinois
ECE 310 **Profs. Liang and Snyder**

Midterm Exam 1

7:00-9:00pm, Wednesday, February 26, 2025

Name: _____

NetID: _____

Score: _____

Problem	Pts.	Score
1	12	
2	12	
3	10	
4	16	
5	18	
6	6	
7	6	
8	20	
Total	100	

Instructions

- You may not use any books, calculators, or notes other than one handwritten two-sided sheet of 8.5" x 11" paper.
- Show all your work to receive full credit for your answers.
- When you are asked to "calculate", "determine", or "find", this means providing closed-form expressions (i.e., without summation or integration signs).
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. No extra booklet or papers will be considered.

(12 Pts.)

1. Mark "**True**" or "**False**" for the following statements.
   Grading: Correct answer = 2 pt.; Incorrect answer = -1 pt. No answer = 0 pts.

   (a) If the output, $y[n]$, of a system is related to its input, $x[n]$, by $y[n] = \sum_{\ell=-\infty}^{\infty} h[\ell]x[n-\ell]$, for some well-defined function $h[n]$, the system must be a causal system. **T/F**

   (b) If the unit pulse response, $h[n]$, of a system $\mathcal{S}$ is absolutely summable, i.e., $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$, then $\mathcal{S}$ must be BIBO-stable regardless if $\mathcal{S}$ is LSI or not. **T/F**

   (c) The z-transform of $x[n] = e^{j\frac{\pi}{4}n}$ does not exist because the ROC is empty. **T/F**

   (d) The parallel connection of two BIBO stable LTI systems always results in another BIBO stable LTI system. **T/F**

   (e) A BIBO stable LTI system has a transfer function given by $H(z) = \frac{1}{1-\sqrt{2}z^{-1}}$. This system must be anti-causal. **T/F**

   (f) An LTI system with a two-sided impulse response is never BIBO stable. **T/F**

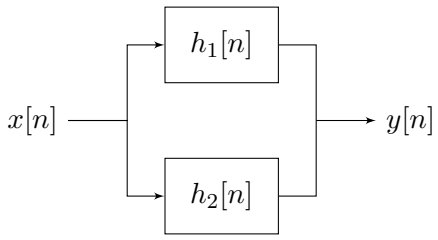
(12 Pts.)

2. For each of the systems with input $x[n]$ and output $y[n]$ shown in the table, indicate by "**yes**" or "**no**" whether the properties indicated apply to the system. **Note**: you do not need to provide proofs/justification. Points will not be deducted for incorrect answers.

	Linear	Shift-Invariant	Causal	Stable
$y[n] = x[n] * j^n u[n]$				
$y[n] = 2x[\lvert n \rvert] + 10$				
$y[n] = e^{x[n]+1}$				

(10 Pts.)

3. Consider the following parallel connection of two LTI systems with impulse responses $h_1[n]$ and $h_2[n]$, respectively.



We are given that $h_1[n] = \delta[n-1]$. Suppose we know the following input-output pair $x[n]$ and $y[n]$.

$$x[n] = \{\underset{\uparrow}{1}, 2, 1\}$$
$$y[n] = \{1, \underset{\uparrow}{2}, 2, 2, 1\}$$

(a) Determine the impulse response $h_2[n]$.

(b) Is the overall system of $h[n]$, i.e. such that $x[n] * h[n] = y[n]$, causal? Justify your reasoning.

(16 Pts.)

4. Compute the response of an LSI system with impulse response $h[n]$ for a given input $x[n]$ in each of the following cases.

   (a) $x[n] = \{1, -2, \underset{\uparrow}{0}, 3, -1, 1\}$, $h[n] = \{\underset{\uparrow}{2}, -1, 3\}$

   (b) $x[n] = \left(-\frac{1}{3}\right)^n u[n]$, $h[n] = \left(\frac{1}{2}\right)^n u[n-3]$

(18 Pts.)

5. Compute the z-transform of each of the following signals including its ROC.

(a) $x[n] = \sum_{k=0}^{2} \left(\frac{1}{2}\right)^k u[n-k]$

(b) $x[n] = 3^n u[-n+2]$

(c) $x[n] = \left(\frac{1}{4}\right)^n \left(\frac{2}{3}\right)^{n-2} u[n-1] + \left(\frac{4}{3}\right)^n u[-n-1]$

(6 Pts.)

6. The transfer function of a causal LSI system is given below:

$$H(z) = \frac{z}{z - \frac{1}{\sqrt{2}}(1+j)}, \quad \text{ROC: } |z| > 1$$

Mark "**True**" or "**False**" for each of the below inputs that will produce an **unbounded** output.

(a) $u[n]$ **T/F**

(b) $e^{j\frac{\pi}{4}n}u[n]$ **T/F**

(c) $e^{-j\frac{\pi}{4}n}u[n]$ **T/F**

(d) $e^{-j\frac{3\pi}{4}n}u[n]$ **T/F**

(e) $\cos(\frac{\pi}{4}n)u[n]$ **T/F**

(f) $4^n u[n]$ **T/F**

(6 Pts.)

7. Consider a causal LSI system given by the following difference equation

$$y[n] = -\frac{3}{2}y[n-1] + y[n-2] + x[n] - \alpha^2 x[n-2],$$

where $\alpha \in \mathbb{C}$. Mark "**True**" or "**False**" for each of the below values for α that will make this system BIBO stable.

(a) 2 **T/F**

(b) -2 **T/F**

(c) j2 **T/F**

(d) -j2 **T/F**

(e) $\frac{\sqrt{2}}{2}$ **T/F**

(f) $\frac{1}{2}$ **T/F**

(20 Pts.)

8. Consider an LTI system with transfer function

$$H(z) = \frac{(2 - 3z^{-1})}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{3}{2}z^{-1}\right)}.$$

(a) Determine the difference equation corresponding to this system.

(b) Determine all the possible outputs $y[n]$ when the input is $x[n] = \delta[n] + \frac{3}{2}\delta[n-1]$.

(c) Determine the ROC of $H(z)$ so that the system is stable and find the corresponding the impulse response $h[n]$.

Scratch paper: Do not remove from the exam!